Year 13 Maths - Further transformations of graphs

Similar questions - Sequences of transformations

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Express $5\cos 2x + 12\sin 2x$ in the form $R\cos(2x-\alpha)$, where $R > 0$ and $0 < \alpha < \frac{1}{2}\pi$, giving α to 4 significant figures. **[3]**

(b) Give full details of a sequence of three transformations that maps the curve $y = \cos x$ onto the curve $y = 5\cos 2x + 12\sin 2x$. **[4]**

(c) Find the greatest and least values of $5\cos 2x + 12\sin 2x$, and the smallest positive value of x at which each occurs. **[4]**

2

(a) Give full details of a sequence of two transformations that maps the graph of $y = \ln x$ onto the graph of $y = \ln(2x-6)$. **[3]**

(b) Show that the equation $\ln(2x-6) = 1 + \ln(x-3)$ has no solution. **[3]**

(c) Solve $\ln(2x-6) = \ln(x+1)$. **[2]**

Question 1

(a) $R = \sqrt{5^2 + 12^2} = 13$, $\tan\alpha = \frac{12}{5}$, so $\alpha = 1.176$ (to 4 s.f.). $5\cos 2x + 12\sin 2x = 13\cos(2x - 1.176)$

(b) Stretch parallel to the x-axis with scale factor $\frac{1}{2}$ (giving $\cos 2x$); translation by $\binom{0.588}{0}$ (giving $\cos(2x-\alpha) = \cos 2\left(x - \frac{\alpha}{2}\right)$); stretch parallel to the y-axis with scale factor 13

(c) Greatest value 13 when $2x - \alpha = 0$, so $x = \frac{\alpha}{2} = 0.588$. Least value -13 when $2x - \alpha = \pi$, so $x = \frac{\alpha+\pi}{2} = 2.16$

Question 2

(a) $\ln(2x-6) = \ln\big(2(x-3)\big)$. Stretch parallel to the x-axis with scale factor $\frac{1}{2}$ (giving $\ln 2x$), then translate by $\binom{3}{0}$ (giving $\ln 2(x-3)$)

(b) $\ln(2x-6) - \ln(x-3) = \ln 2$, so the equation would need $\ln 2 = 1$. But $\ln 2 = 0.693\ldots \neq 1$, so there is no solution

(c) $2x - 6 = x + 1$, so $x = 7$. Check: $\ln 8 = \ln 8$ ✓